

Solution

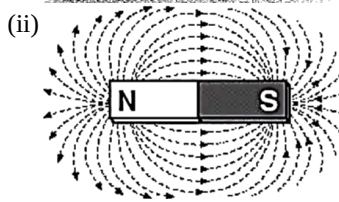
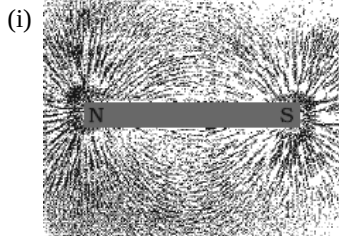
MAGNETISM -01

Class 10 - Science

Section A

1. Read the text carefully and answer the questions:

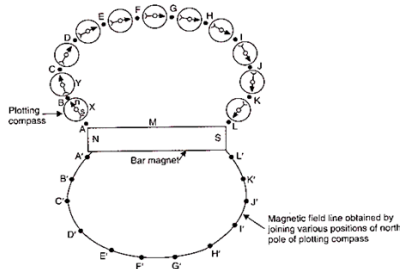
A student fixes a sheet of white paper on a drawing board using some adhesive materials. She places a bar magnet in the centre of it and sprinkles some iron filings uniformly around the bar magnet using a salt-sprinkler. On tapping the board gently, she observes that the iron filings have arranged themselves in a particular pattern.



(iii) The direction of a magnetic field at a point is determined by placing a small compass needle. The N - pole of compass indicates the direction of magnetic field at that point.

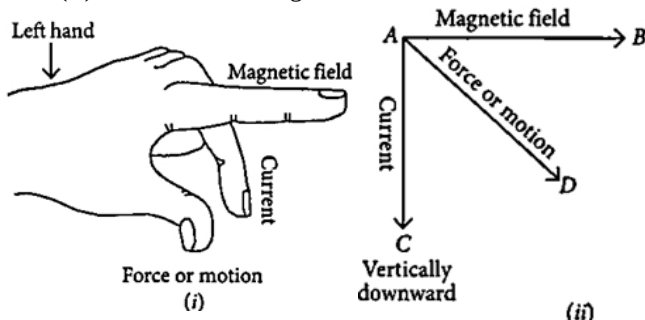
If magnetic field lines intersect each other, then at the intersection point there will be two directions of the same field which is not possible.

(iv) Magnetic field lines can be drawn by moving a small compass from point to point around a magnet. At each point, draw a short line in the direction of the compass needle. Joining the points together reveals the path of the magnetic field lines.



2. Read the text carefully and answer the questions:

Andre Marie Ampere suggested that a magnet must exert an equal and opposite force on a current-carrying conductor, which was experimentally found to be true. But we know that current is due to charges in motion. Thus, it is clear that a charge moving in a magnetic field experience a force, except when it is moving in a direction parallel to it. If the direction of motion is perpendicular to the direction of magnetic field, the magnitude of force experienced depends on the charge, velocity (v), strength of magnetic field (B), and sine of the angle between v and B . Direction of magnetic force is given by Fleming's left-hand rule.



(i) Fleming's left-hand rule is used to determine the direction of force on electron i.e., in south direction.

(ii) Force = $q (\mathbf{V} \times \mathbf{B}) = qVB \sin\theta$

Where, θ is angle between velocity and magnetic field.

So, $\sin\theta$ is maximum when θ is 90°

or velocity is perpendicular to magnetic field.

(iii) As the direction of current is taken opposite to the direction of motion of electrons, therefore, current from the motion of electron and proton is in the same direction, i.e., from bottom to top. Now, according to Fleming's left-hand rule, the electron and the proton experience forces both pointing into the plane of paper.

(iv) We know that both the directions are perpendicular, thus for force direction = ?

Using Fleming's left-hand rule,

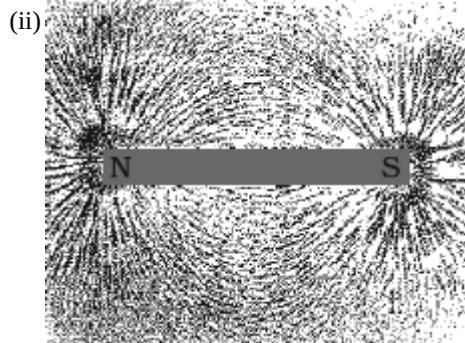
Direction of force is perpendicular to the direction of magnetic field and current.

Thus direction of force is opposite to electron motion into the page at 90°

3. Read the text carefully and answer the questions:

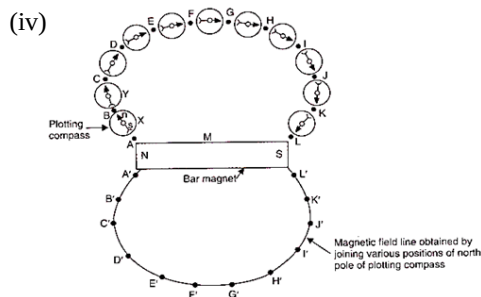
A student fixes a sheet of white paper on a drawing board using some adhesive materials. She places a bar magnet in the centre of it and sprinkles some iron filings uniformly around the bar magnet using a salt sprinkler. On tapping the board gently, she observes that the iron filings have arranged themselves in a particular pattern.

(i) The bar magnet kept at the centre of board has its magnetic field around it. The iron filings sprinkled on the board experience a force on them due to the magnetic field of bar magnet. So, when the student taps the board the iron filings align themselves according to the magnetic field lines of the bar magnet.



(iii) The direction of a magnetic field at a point is determined by placing a small compass needle. The N - pole of compass indicates the direction of magnetic field at that point.

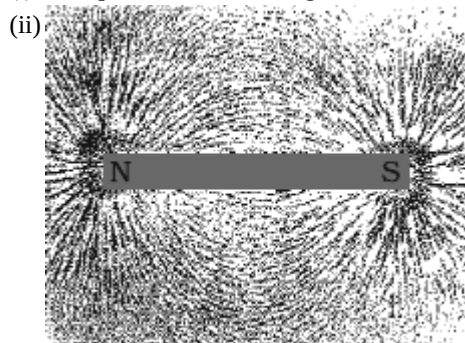
Two magnetic field lines do not intersect each other because if there was point of intersection, The compass needle would point towards 2 directions.



4. Read the text carefully and answer the questions:

A student fixes a sheet of white paper on a drawing board using some adhesive materials. She places a bar magnet in the centre of it and sprinkles some iron filings uniformly around the bar magnet using a salt sprinkler. On tapping the board gently, she observes that the iron filings have arranged themselves in a particular pattern.

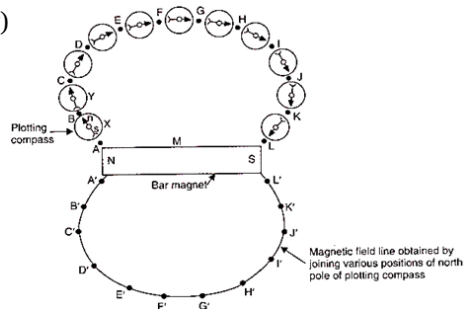
(i) This pattern of iron fillings demonstrate the magnetic field lines.



(iii) The direction of a magnetic field at a point is determined by placing a small compass needle. The N - pole of compass indicates the direction of magnetic field at that point.

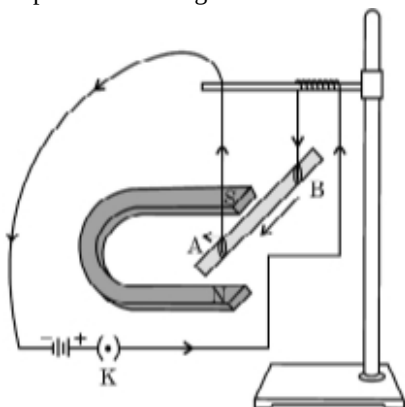
Two magnetic field lines do not intersect each other because if there was point of intersection, The compass needle would point towards 2 directions.

(iv)



5. Read the text carefully and answer the questions:

A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminum rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions:

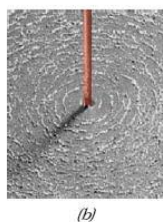
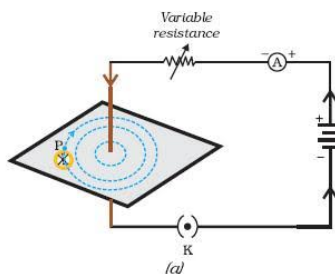


- (i)
 - i. If the polarity of the magnet and the direction of current both are reversed, using Fleming's left hand rule it gets displaced towards the left.
 - ii. Devices that use current-carrying conductors and magnetic fields are electric motor, electric generator etc.
- (ii) When a current-carrying conductor is placed in a magnetic field, it experiences a force, due to which the rod gets displaced.
- (iii) The rule that determines the direction of the force on the conductor AB is Fleming's left-hand rule.

According to Fleming's left-hand rule, stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular.

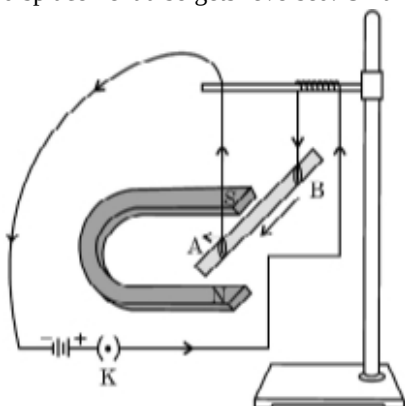
If the first finger points in the direction of the magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or force.

- (iv) The magnetic field lines around a current carrying conductor can be represented by concentric circles which can be determined by right hand thumb rule.



6. Read the text carefully and answer the questions:

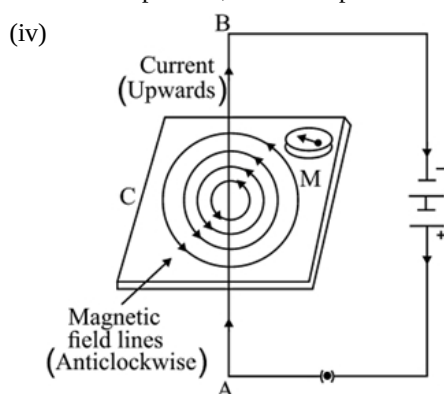
A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminum rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions:



- (i) When current is passed through the rod it produces magnetic field due to which it feels a force and due to it rod displaces from its position.
- (ii) Fleming's left hand rule gives the direction of force experienced by current carrying conductor placed in an external magnetic field.

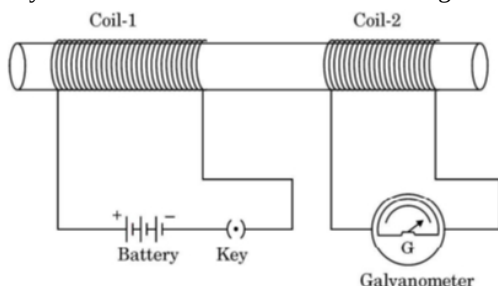
According to Fleming's left hand rule if we arrange our thumb, forefinger and middle finger of the left hand perpendicular to each other, then the thumb points towards the direction of the magnetic force, the forefinger points towards the direction of the magnetic field and the middle finger points towards the direction of the current.

- (iii) a. According to Fleming's left-hand rule, the rod will get displaced upwards.
b. Devices that use current-carrying conductors and magnetic fields are electric motors, electric generators, loudspeakers, and microphones.



7. Read the text carefully and answer the questions:

Take two different coils of copper wire having large number of turns, say 50 and 100 turns respectively. Insert them over a non-conducting roll as shown in the given figure. Connect the Coil-1, having large number of turns, in series with a battery and a plug key. Also connect the other Coil-2 with a galvanometer.



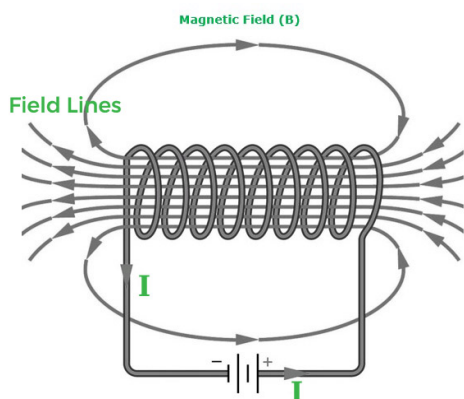
- (i) The reason involved is electromagnetic induction.
When current is passed through coil 1 magnetic field changes in coil 2 due to which an induced current starts flowing in coil 2 and the galvanometer present in coil 2 shows some deflection.
- (ii) The electromagnetic induction is the phenomenon involved above.
The process by which a changing magnetic field in a conductor induces a current in another conductor is called electromagnetic induction.
- (iii) 1. A momentary deflection is shown by the galvanometer.

2. A momentary deflection is shown by the galvanometer but in the opposite direction.

- (iv) i. When bar magnet is pushed into the coil there is some deflection in the galvanometer.
ii. When bar magnet is held stationary there is no change in magnetic field due to which there is no change in the galvanometer.

8. Read the text carefully and answer the questions:

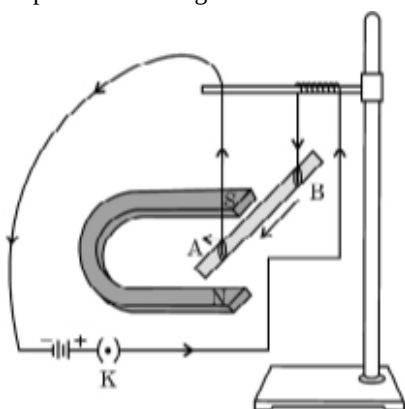
An insulated copper wire wound on a cylindrical cardboard tube such that its length is greater than its diameter is called a solenoid. When an electric current is passed through the solenoid, it produces a magnetic field around it. The magnetic field produced by a current-carrying solenoid is similar to the magnetic field produced by a bar magnet. The field lines inside the solenoid are in the form of parallel straight lines. The strong magnetic field produced inside a current-carrying solenoid can be used to magnetize a piece of a magnetic material like soft iron when placed inside the solenoid. The strength of the magnetic field produced by a current-carrying solenoid is directly proportional to the number of turns and strength of the current in the solenoid.



- (i) Magnetic field inside the infinite solenoid is uniform. Hence it is the same at all points.
(ii) The end of the current carrying solenoid at which the current flows anti-clockwise behaves as a north pole while that end at which the direction of current clockwise behaves as a south pole and this is according to clock wise.
(iii) For a long solenoid, magnetic field $B \propto In$; where I is the flowing current and n is number of turns per unit length in the solenoid. Therefore, in the given case magnetic field will remain unchanged.
(iv) For a solenoid, if we imagine gripping the solenoid with your right hand so that your curl fingers follow the direction of the current then your thumb will point towards the north end of the electromagnet.

9. Read the text carefully and answer the questions:

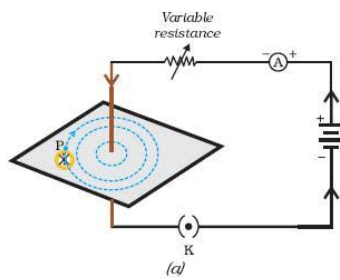
A student was asked to perform an experiment to study the force on a current carrying conductor in a magnetic field. He took a small aluminum rod AB, a strong horse shoe magnet, some connecting wires, a battery and a switch and connected them as shown. He observed that on passing current, the rod gets displaced. On reversing the direction of current, the direction of displacement also gets reversed. On the basis of your understanding of this phenomenon, answer the following questions :



- (i) The displacement of the conductor is maximum when the direction of the current is at right angles to the direction of the magnetic field.
(ii) The rule that determines the direction of the force on the conductor AB is Fleming's left-hand rule.
According to Fleming's left-hand rule, stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular.
If the first finger points in the direction of the magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or force.
(iii) i. According to Fleming's left-hand rule, the rod will get displaced upwards.

- ii. Devices that use current-carrying conductors and magnetic fields are electric motors, electric generators, loudspeakers, microphones, etc.

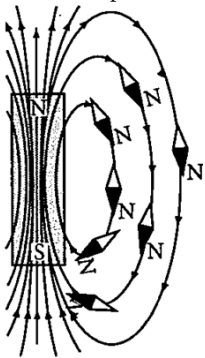
(iv)



(b)

10. Read the text carefully and answer the questions:

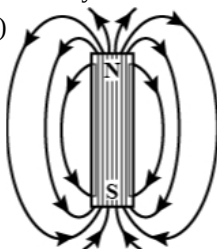
A magnetic field is described by drawing the magnetic field lines. When a small north magnetic pole is placed in the magnetic field created by a magnet, it will experience a force. And if the north pole is free, it will move under the influence of the magnetic field. The path traced by a north magnetic pole free to move under the influence of a magnetic field is called a magnetic field line.



Since the direction of the magnetic field line is the direction of the force on a north pole, so the magnetic field lines always begin from the N-pole of a magnet and end on the S-pole of the magnet. Inside the magnet, however, the direction of magnetic field lines is from the S-pole of the magnet to the N-pole of the magnet. Thus, the magnetic field lines are closed curves. When a small compass is moved along a magnetic field line, the compass needle always sets itself along the line tangential to it. So, a line drawn from the south pole of the compass needle to its north pole indicates the direction of the magnetic field at that point.

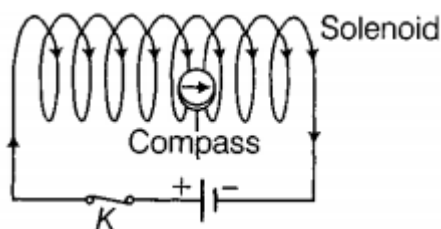
- (i) No two magnetic field lines are found to cross each other. If two field lines crossed each other, it would mean that at the point of intersection, the compass needle would point in two directions at the same time, which is not possible.
- (ii) The magnetic field and hence the magnetic line of force exist in all the planes all around the magnet.
- (iii) The magnetic lines of force are uniform and strong at point C and they diverge as they move towards points A and B and the distance between the lines increases. Eventually, the strength of the magnetic field is strong where the lines are closer and they weaken as the closeness decreases i.e., at point C.

(iv)



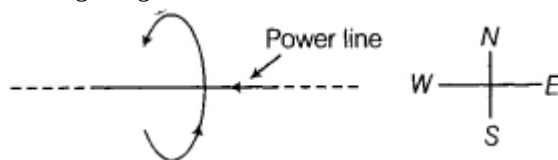
Section B

11. i.



ii. The direction of the magnetic field inside the solenoid always points from the induced South pole towards the induced North pole.

12. i. According to right-hand thumb rule,

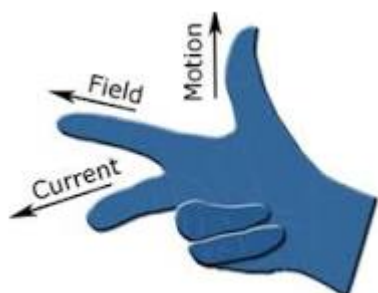


a. The direction of magnetic field at a point above the power line. is from South to North.

b. The direction of magnetic field at a point below the power line is from North to South.

ii. a. If the current in the coil X is changed then some current will definitely be induced in the coil Y. When the current in coil X is changed, the magnetic field associated with it also changes. As a result, the magnetic field around coil Y also changes. This change in magnetic field lines around coil Y induces an electric current in it. This process is known electromagnetic induction.

b. **Fleming's Right-hand Rule:** According to Fleming right-hand rule, Adjust your right-hand thumb, forefinger and middle finger in such a way that all are perpendicular to one another. If forefinger shows the direction of magnetic field and Thumb shows the direction of motion of the conductor then middle finger gives the direction of current.



13. According to Right Hand Thumb Rule, imagine the straight conductor in your right hand such that the thumb points in the direction of the current, then the direction of curling of fingers of the right hand gives the direction of magnetic field lines. At P, the direction of magnetic field is towards the paper and at point B, the direction of magnetic field is out of the paper. Since $r_1 > r_2$, so the strength of magnetic field at P is less than that at Q.

14. i. The constant current flowing in the straight wire produces a constant magnetic field. Hence, no induced current is produced in the loop.

ii. Since current in the straight wire is changing, hence, induced current will be produced in clockwise direction.

15. i. a. Right hand thumb rule will determine the direction of a magnetic field around a straight conductor carrying current.

b. Fleming's left had rule will determine the direction of a force experienced by a current-carrying straight conductor placed in a magnetic field which is perpendicular to it.

c. Fleming's right hand rule will determine the direction of current induced in a coil due to its rotation in a magnetic field.

ii. a. The **right-hand thumb** rule is used to find the directions of the magnetic field around a current-carrying straight conductor.

b. **Fleming's left-hand rule** (motor rule): helps in understanding the direction of the magnetic force acting on a conductor.

c. **Fleming's right-hand rule** (dynamo rule) helps in understanding the direction of induced current when a conductor moves in a magnetic field.

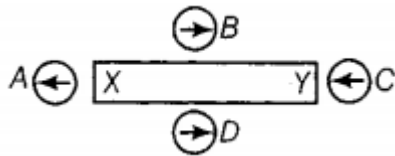
16. i. The magnetic field will be acting in vertically downward direction in accordance with Fleming's left-hand rule. [Direction of the current should be considered in a direction opposite to the direction in which the electrons move].

ii. Current will definitely be induced in coil B if current is changed in coil A. If current in A is increased, the induced current in B will be in direction opposite to that in A. If current in A is decreased, induced current in B will be in same direction as in A.

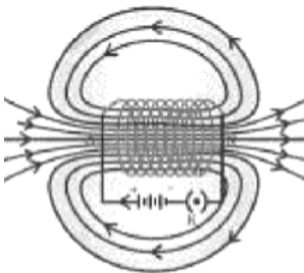
This change in magnetic field lines around coil B induces an electric current in it. This process is known electromagnetic induction.

17.
 - i. Relative closeness of field lines indicates the strength of magnetic field. Since field lines are crowded around the ends of the solenoid, hence these are the regions of strongest magnetism.
 - ii. The direction of the field will also be reversed.
18.
 - i. The magnitude of induced current increases.
 - ii. The induced current is reversed.
19. As we know, force acting on current carrying conductor is given by the relation:
 $F = BIL$.
Here B is strength of the magnetic field, I is the current flowing in conductor and L is the length of the conductor.
 - i. When I increase, F also increases hence displacement of the rod increases.
 - ii. When a stronger horse shoe magnet is inserted, magnetic field B increases. So force F increase. Hence displacement increases.
20.
 - i. The encircled cross represents the direction of the $\vec{v}$ electron. Applying the right-hand rule for the cross product of vectors, we find that the direction of the vector $(\vec{V} \times \vec{B})$ is towards east since $\vec{F}_m = -e(\vec{V} \times \vec{B})$, therefore the force is directed towards the west.
 - ii. As the length of the wire is parallel to the wire, the cross product of $\vec{L} \times \vec{B}$ is $(LB \sin 0) = 0$
21.
 - i. Due to the interaction between the magnetic fields of the compass needle and the magnetic field of a bar magnet, a magnetic compass needle shows deflection when a bar magnet is brought close to it. The various features of magnetic field lines are:
 - a. The direction of magnetic field lines is from North to South pole.
 - b. Magnetic field lines are closed curves.
 - c. The more the lines are closed, the more will be the magnetic field.
 - d. No two field lines can cross each other.
 - e. Strength of magnetic field near the poles.
 - ii. The activity to demonstrate that a current-carrying conductor experiences a force perpendicular to its length and the external magnetic field can be explained as follows:
Activity: To show the effect of magnetic field on a current-carrying conductor
Materials Required: For this, we need to take a small aluminum rod, a horseshoe magnet, battery, plug key, wires, and a stand.
 - a. Suspend an aluminum rod horizontally from the stand and two wires at the ends of it are tied. The wires are connected to a Rheostat, battery and a key so that a circuit is completed,
 - b. Place a horseshoe magnet in such a manner that the aluminum rod is between the poles of a magnet.Assume that the above the aluminum rod is South pole of the magnet and below, the north pole of the magnet. Insert the plug key and current is supplied to the rod.
Observation: the aluminum rod is deflected towards the left direction
On changing the direction of the current, the rod is deflection in the right direction.
Hence, it demonstrates that a current-carrying conductor experiences a force perpendicular to its length and the external magnetic field
The direction of the magnetic field can find out with the help of Fleming's left-hand rule. Let current is moving in an anticlockwise direction, then the direction of the magnetic field will be in clockwise direction i.e. at the top of the loop whereas vice-versa in case of the clockwise direction of the current.
22.
 - i. The magnitude of induced current increases.
 - ii. The induced current is zero.
23.
 - i. The current is in the east-west direction. By applying the right-hand thumb rule, we get that the magnetic field (at any point below or above the wire) that turns clockwise in a plane perpendicular to the wire, when viewed from the east end, and anti-clockwise, when viewed from the west end.
 - ii. Following are the methods of producing magnetic fields:
 - a. By using a permanent magnet we can produce a magnetic field and it can be visualized by spreading iron fillings on a white paper and keeping a magnet beneath the paper.
 - b. A current-carrying straight conductor produces a magnetic field.
24.
 - i. N-pole
 - ii. Clock face rule

25. i. When the coil is kept vertically in North-South plane and the current is flowing in the anti-clockwise direction through the loop, then the magnetic field is in the East-West direction.
 ii. When the coil is kept vertically in East-West plane and current through the coil is in the anti-clockwise direction, then the magnetic field is in the South-North direction.
 iii. In this case, the direction of the field for the observer positioned below the coil is in the downward direction.
26. i.



- ii. The North Pole is X.
27. A coil of insulated copper wire wound in the form of a cylinder is a solenoid. When current is passed through a solenoid, it produces magnetic field lines like a bar magnet. The pattern of magnetic field lines of a solenoid is shown in the figure. Inside the solenoid, field lines are parallel to each other. This indicates that the magnetic field is uniform.



28. i. Strength of the magnetic field falls as distance increases. This is indicated by the decrease in degree of closeness of the lines of field.
 ii. Current-carrying loops behave like bar magnets and both have their associated field lines. This modifies the already existing earth's magnetic field and a deflection results.
29. Strength of the magnetic field falls as distance increases. This is indicated by the decrease in degree of closeness of the lines of field.

Suppose x number of such bulbs can be used.

Power of one bulb = 100 watt

Power of x bulbs (p) = 100x watts

Potential difference (V) = 220 Volts

I = 5 ampere

We know, $P = VI$

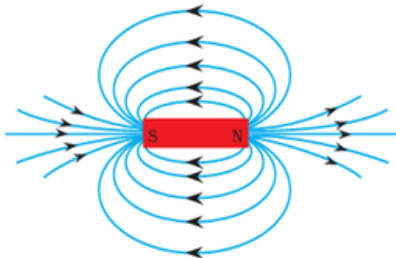
$$100x = 220 \times 5$$

$$\text{Therefore, } x = \frac{220 \times 5}{100}$$

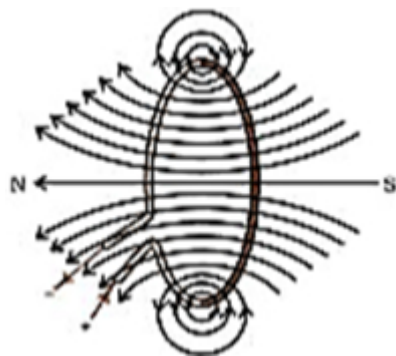
= 11 bulbs

30. Uniform magnetic field: The space or region where field is same everywhere is known as Uniform magnetic field.

Non-uniform magnetic field: The magnetic field which is unequal in magnitude and direction at every point in the space is called non-uniform magnetic field.



31. a.



b. Right Hand Thumb Rule:- When a current carrying straight conductor is held in the right hand in such a way that the thumb points towards the direction of the current, then the fingers will wrap around the conductor in the direction of the field lines of the magnetic field

32. i. There must be a relative motion between the coil of wire and a magnet.

ii. Resistivity of coil will determine the resistance of the coil and induced current through it, as induced current = $\frac{\text{emf}}{\text{resistance}}$

33. i. The direction of force acting on the electron beam will be into the page.

ii. The angle between velocity and magnetic field is zero. Therefore, magnetic force on the particle is zero.

34. i. Since force acting on the rod is directly proportional to the current passing through it. Therefore, the displacement will be increased when current is increased.

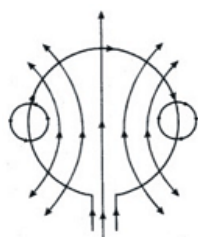
ii. Stronger the magnet, more will be the force and hence the displacement.

35. i. Electrical energy consumed in three hours = $P \times t = 2 \text{ kW} \times 5 \text{ hour} = 10 \text{ kWh}$

Unit cost = ₹ 4 per kWh

Therefore, the cost of energy used for three hours = $4 \times 10 = ₹ 40$

ii. The direction of the magnetic field inside and outside is loop as shown.



In the following diagram the current is flowing clockwise. If we are applying right-hand thumb rule to the left side of the loop then the direction of magnetic field lines inside the loop are going into the table while outside the loop they are coming out of the table. If we are applying right-hand thumb rule to the right side of the loop then the direction of magnetic field lines inside the loop are again going into the table while outside the loop they are coming out of the table.

36. A magnetic compass can be used to detect the presence of the magnetic field around the wire.

Maxwell's right-hand thumb rule is used to predict the direction of magnetic field. It states that " if you hold a current carrying conductor in right hand, such that the thumb points in the direction of electric current, then the direction in which fingers encircle, gives the direction of magnetic field".

37. When the magnetic field and direction of electric current are in the same plane, there would be deflection in the compass. For maximum deflection, magnetic field and direction of current should be in mutually perpendicular plane. So, for no deflection, conductors should be kept parallel to the magnetic compass. For maximum deflection, it should be placed perpendicular to the compass.

38. i. The terminal to be connected by green wire is the earthing terminal.

ii. Mahesh needs to be more careful, concerned about safety measures and be less ignorant about simple safety practices.

iii. I would have checked all other devices to ensure the further safety.

39. i. The direction of the magnetic field at a point can be found by placing a small magnetic compass at that point. The north end of the needle of a compass indicates the direction of magnetic field at a point where it is placed.

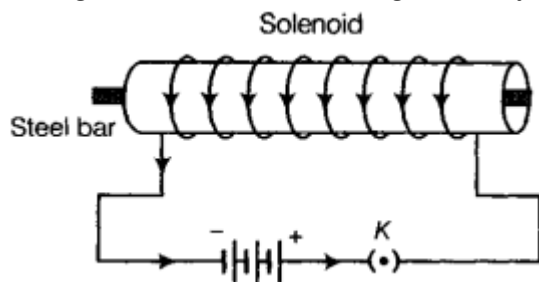
ii. The direction of magnetic field at the centre of a current-carrying circular loop is perpendicular to the plane of the loop.

40. i. The magnetic field and hence the magnetic line of force exist in all the planes all around the magnet.

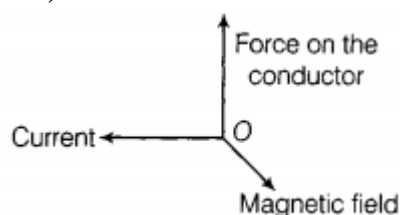
ii. The relative strength of the magnetic field is shown by the degree of closeness of the field lines and the direction of the magnetic field is obtained by tangent to the field lines at the point of intersect.

41. The conditions to obtain permanent electromagnet, if a current carrying solenoid is used, are given as follows:

- i. The current through the solenoid should be direct current.
- ii. The rod inside the solenoid must be made of a magnetic material.
- iii. The magnetic material should be of high retentivity.



42. i. The direction of magnetic field is perpendicular to both current and force on the conductor. (according to Fleming's left-hand rule).



- ii. a. As the distance from conductor increases, the strength of the magnetic field decreases i.e. $B \propto \frac{1}{r}$.
 - b. As the magnitude of the current flowing through the conductor increases, the strength of magnetic field increases, i.e. $B \propto I$.
43. i. The magnetic field lines of force are the lines drawn in a magnetic field along which a hypothetical north magnetic pole would move if it is free to do so.

The direction of a magnetic field at a point is the direction of the resultant force acting on a hypothetical north pole placed at that point. The tangent at any point on the magnetic field line gives the direction of magnetic field at that point. The direction of the magnetic field at a point can be found by placing a small magnetic compass at that point. The north end of the needle indicates the direction of the field.

Two important properties of the magnetic lines of force are:>

- a. The magnetic lines of force start from north pole and terminate at south pole. Inside the magnet they travel from south pole to north pole. Thus, they are closed curves.
- b. They do not intersect each other because at the point of intersection there will be two directions of same magnetic field which is impossible.

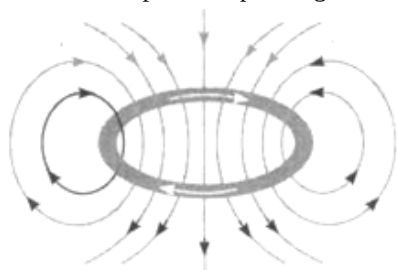
- ii. Power $P = 2\text{kW} = 2,000\text{ W}$, $V = 220\text{ volts}$, $I = ?$

$$P = VI \text{ or } I = \frac{P}{V}$$

$$I = \frac{2000}{220} = \frac{100}{11} = 9.09\text{ A}$$

A current of 9.09 A will flow in the circuit. Since the current rating of circuit is 5A, the fuse (of 5A) rating if inserted in circuit will burn up. If no fuse has been put in the circuit, there may be a fire.

44. i. Magnetic field inside infinite solenoid is uniform. Hence it is same at all points.
- ii. For a long solenoid, magnetic field $B \propto In$; where I is the flowing current and n is number of turns per unit length in the solenoid. Therefore, in the given case magnetic field will remain unchanged.
45. i. Since the current passes through the loop in clockwise direction, therefore, the front face of the loop will be the south pole and the back face, i.e., the face touching the table will be north pole. According to right-hand rule, the direction of magnetic field inside the loop will be pointing downward. Outside the loop, the direction of the magnetic field will be upward.



- ii. At the axis of a current-carrying circular loop, magnetic field lines are straight. Hence they appear closer to each other. When we move towards the periphery of current carrying circular loop, magnetic field lines turn so as to encircle the conductor. Due

to this, they appear to be diverging.